CODIMENSION ZERO POINTS DETECTED BY GENERIC POINTS OF COMPONENTS ON ETALE NEIGHBOURHOODS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original reference 0BA9: sources/stacks-chow/tags.txt, source lines 14202–14202. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 04NA: sources/stacks-chow/tags.txt, source lines 6071–6071. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme and let X be an algebraic space over S. Let $x \in |X|$. Consider étale morphisms $a : U \to X$ where U is a scheme. The following are equivalent*

1. *x is a point of codimension 0 on X,*
2. *for some $U \to X$ as above and $u \in U$ with $a(u) = x$, the point u is the generic point of an irreducible component of U, and*
3. *for any $U \to X$ as above and any $u \in U$ mapping to x, the point u is the generic point of an irreducible component of U.*

If X is representable, this is equivalent to x being a generic point of an irreducible component of $|X|$.

Proof. Observe that a point u of a scheme U is a generic point of an irreducible component of U if and only if $\dim(\mathcal{O}_{U,u}) = 0$ (Properties, Lemma 0BA9). Hence this follows from the definition of the codimension of a point on X (Definition 04NA). □

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